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Chapter 3

Forecasting

Operations Management

FOURTEENTH EDITION

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Chapter 3: Learning Objectives

You should be able to:

- List features common to all forecasts
- Explain why forecasts are generally wrong
- List the elements of a good forecast
- Outline the steps in the forecasting process
- Use a **naïve method** to make a forecast
- Prepare a **moving average forecast**
- Prepare a **weighted-average forecast**
- Prepare an **exponential smoothing forecast**
- Prepare a **linear trend forecast**
- Compute and use **seasonal relatives**
- Compute and monitor **forecast errors**

Forecast

- Forecast is **a statement about the future value of a variable** of interest.
- Manager make forecasts for about such things like:
 - *budgeting, profit, demand, resource availability, capacity, sales, production, inventory, personnel, purchasing*, and more.
- Forecasts are **made with reference to a specific *time horizon***, it may be fairly short or somewhat longer.

Two Important Aspects of Forecasts

Expected level of demand

- The level of demand may be a function of some structural variation, such as '**trend**' or '**seasonal**' variation

Accuracy of forecast

- The ability of forecasters to correctly **model demand**, **random variation**, and sometimes **unforeseen events**.
- Related to the potential size of *forecast error*.

Forecasts in Business Organizations

- **Accounting.** New product/process cost estimates, *profit* projections, cash management.
- **Finance.** Equipment replacement needs, timing and amount of *funding / borrowing* needs.
- **Human resources.** *Hiring* activities, *recruitment*, *interviewing*, *training*, *layoff*.
- **Marketing.** *Pricing* and *promotion*, *e-business* strategies, global *competition* strategies.
- **Operations.** *Schedules*, *capacity* planning, *work assignments* and *workloads*, *inventory* planning, *make-or-buy* decisions, *outsourcing*.
- **Product/service design.** *Revision* of current features, *design* of new products or services.

Forecast Uses

Plan the system (System planning decisions)

- Generally **involves long-range plans** related to:
 - Facility location
 - Facility and equipment levels
 - Types of products and services to offer

Plan the use of the system (System use decisions)

- Generally **involves short- and medium-range plans** related to:
 - Scheduling
 - Inventory levels
 - Workforce levels
 - Budgeting
 - Purchasing
 - Production

Elements of a Good Forecast

A good forecast...

- Should be *timely*
- Should be *accurate*
- Should be *reliable*
- Should be expressed in *meaningful units*
- Should be *in writing*
- Technique should be *simple to understand and use*
- Should be *cost-effective*

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Forecasting and the Supply Chain

- **Accurate forecasts are vital for the supply chain, inaccurate forecasting for:**
 - Shortages of materials, part, and services lead to:
 - Increased costs
 - Missed deliveries
 - Work disruption
 - Poor customer service
 - Excess of materials, services, and/or capacity lead to:
 - Increased costs
 - Waste of resource
 - Poor customer service

Steps in the Forecasting Process

1. Determine the **purpose** of the forecast



2. Establish a **time horizon**



3. Obtain, clean, and analyze **appropriate data**



4. Select a **forecasting technique**



5. Make the **forecast**



6. Monitor the **forecast errors**

Forecast Accuracy and Errors

- Forecast should account for errors
 - It is important to provide an indication of the extent to which the forecast might deviate from the value of the variable that actually occurs.
- **Forecast errors** should be monitored
 - **Error = Actual – Forecast**
 - If errors fall beyond acceptable bounds, corrective action may be necessary.

Forecasting Approaches

Qualitative forecasting

- permit the inclusion of *soft* information, such as:
 - Human factors
 - Personal opinions
 - Hunches
 - ...

Quantitative forecasting

- rely on *hard* data, such as:
 - Capacities of machine
 - Operating hours of Workers
 - Stock level
 - ...

Forecasting Techniques

- **Judgmental forecast**

- Rely on *subjective inputs*, and provide insights that are not otherwise available.

- **Time-series forecast**

- Use *historical data* as the assumption, and infer that the future will be like the past.

- **Associative forecast**

- Use *equations* that consist of one more *explanatory variables* that can be used to predict demand.

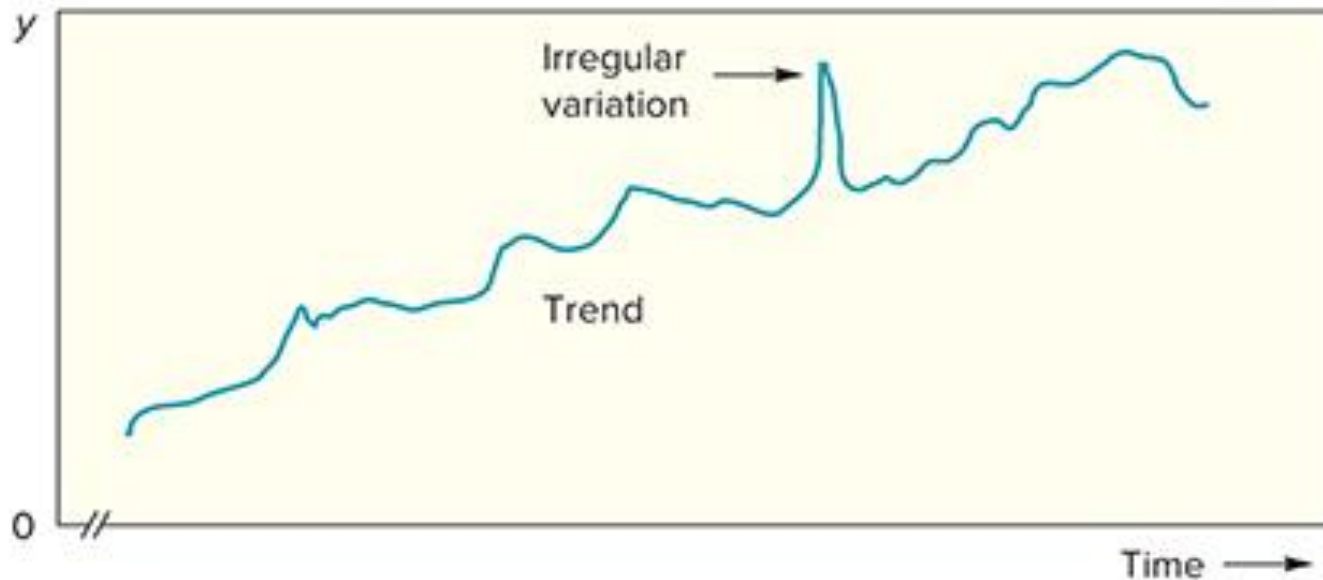
Forecasts on Time-series Data

- **Time-series** – a *time-ordered sequence* of observations taken at regular time intervals.
- Data of time-series may be measurements of *demand*, *cost*, *profit*, etc. It include one or more influence factors, such as:
 - **Trend**
 - **Seasonality**
 - **Cycles**
 - **Irregular variations**
 - **Random variation**

Trends, Seasonality and Cycle

Trend

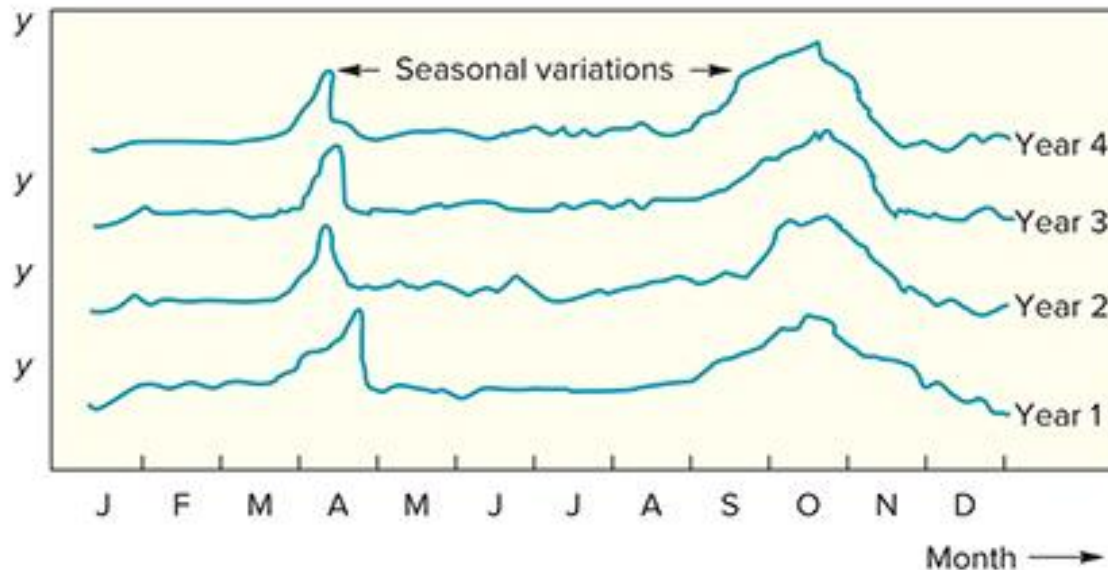
- A long-term, *upward or downward movement* in data.
 - Population shifts
 - Changing income



Trends, Seasonality and Cycle

Seasonality

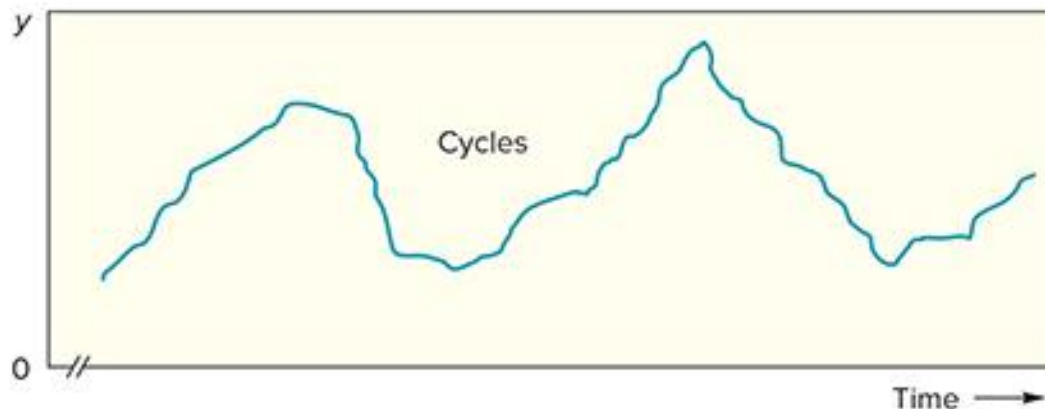
- A short-term, *fairly regular* variations related to the calendar or time of day.
 - Restaurant
 - Playground



Trends, Seasonality and Cycle

Cycle

- **Wavelike** variations lasting **more than one year**, and often related to:
 - Economic
 - Political
 - Production conditions



Cycles and Variations

Irregular Variation

- Due to *unusual circumstances* that do not reflect typical behavior.
 - Labor strike
 - Weather event

Random Variation

- *Residual variation* that remains after all other behaviors have been accounted for.
 - Epidemic

Time-Series Forecasting Approaches

- **Naïve forecast**
- **Techniques for Averaging**
 - Moving average
 - Weighted moving average
 - Exponential smoothing
- **Techniques for Trend**
 - Linear Trend Equation
- **Techniques for Seasonality**

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Naïve forecast

- Uses a *single previous value* of a time series as the basis for a forecast.
- That is, **forecast for a time period is equal to previous time period's value.**
- Can be used with:
 - **A stable time series**
 - **Seasonal variations**
 - **Trend**

Naïve forecast

- For data with a **stable series**...
 - If demand for a product last week was 20 cases, the forecast for this week is 20 cases.
 - The forecast for demand for turkeys this Thanksgiving season is equal to demand for turkeys last Thanksgiving.
 - The forecast for highway traffic volume this Friday is equal to the highway traffic volume last Friday.

Naïve forecast

- **Practice:**

- if demand for a component that the last two periods were 50 and 53. Applying Naïve forecast, the next period forecast would be?

Period	Actual	Change from Previous Value	Forecast
1	50		
2	53		
3	52		
4			?

Thinking:

if actual demand for a component the last three periods were 50, 53, and 52, the next period forecast would be...?

Naïve forecast

- **Advantages**
 - virtually no cost
 - quick and easy to prepare data
 - easily understandable
- **Disadvantages**
 - inability to provide highly accurate forecasts

Averaging techniques

- Averaging techniques can work best **when a series tends to vary around an *average*.**
- These techniques generate forecasts that **reflect recent values of a time series.**
- Averaging techniques include:
 - 1. Moving average**
 - 2. Weighted moving average**
 - 3. Exponential smoothing**

Moving Average₁

Use an *average of consecutive periods* of data in generating a forecast.

- **Equation:**

$$F_t = \frac{\sum_{i=1}^n A_{t-i}}{n} = \frac{A_{t-n} + \cdots + A_{t-2} + A_{t-1}}{n}$$

where

F_t = Forecast for time period t

A_{t-i} = Actual value in period $t - i$

n = Number of periods in the moving average

Moving Average₂

- **Practice:**

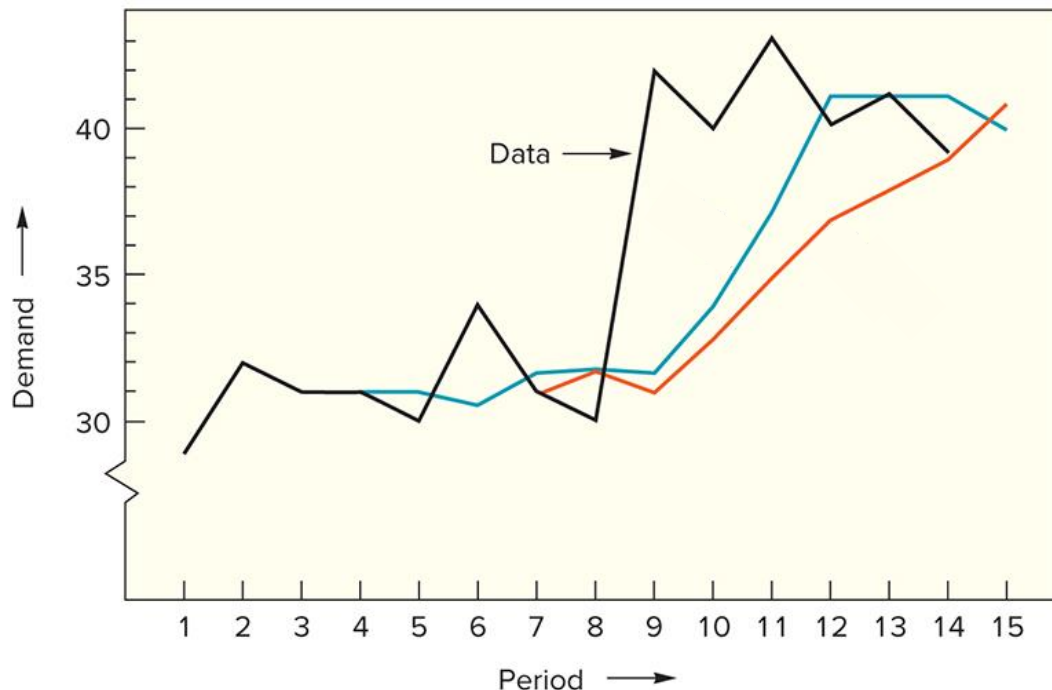
- a) Compute a three-period moving average forecast given demand for shopping carts for the last five periods.
- b) If actual demand in period 6 turns out to be **38**, compute a three-period moving average forecast for period 7.

Period	Demand	
1	42	
2	40	
3	43	} the 3 most recent demands
4	40	
5	41	
6	38	

Moving Average₃

The forecast will “*moves*” base on the most recent values to reflect the change in value.

The **model sensitivity** is determined by the number of periods used in calculating the average.



※

MA₃ represent using **3 periods** for forecasting;

MA₅ represent using **5 periods**.

Moving Average₄

- **Advantage**
 - easy to compute and understand.
- **Disadvantage**
 - all values are weighted equally.
 - need to consider suitable number of periods.

Weighted Moving Average₁

The *most recent values* in a time series are given more **weight** in forecasting, and the weights must sum to 1.0.

- **Equation:**

$$F_t = w_{t-1}(A_{t-1}) + w_{t-2}(A_{t-2}) + \cdots + w_{t-n}(A_{t-n})$$

where

F_t = Forecast for time period t

w_{t-n} = weight for period $t - 1, t - 2, \dots t - n$, etc.

A_{t-n} = the actual value for period $t - 1, t - 2, \dots t - n$, etc.

n = Number of periods in the weighted moving average

Weighted Moving Average₂

- **Practice:**

- Compute a weighted average forecast using a weight of **0.40** for the most recent period, **0.30** for the next most recent, **0.20** for the next, and **0.10** for the next.
- If the actual demand for period 6 is **39**, forecast demand for period 7 using the same weights as in part a).

Period	Demand	
1	42	
2	40	} Only the <i>four most recent</i> demands are used
3	43	
4	40	
5	41	
6	39	

Weighted Moving Average₂

- Solution:***

a) $F_6 = 0.4*(41) + 0.3*(40) + 0.2*(43) + 0.1*(40) = 41.0$

b) $F_7 = 0.4*(39) + 0.3*(41) + 0.2*(40) + 0.1*(43) = 40.2$

Period	Demand	
1	42	
2	40	} Demands are used to forecast F_6
3	43	
4	40	
5	41	
6	39	} Demands are used to forecast F_7

Weighted Moving Average₃

- **Advantage**
 - more reflective of the most recent occurrences.
- **Disadvantage**
 - the choice of weights is somewhat arbitrary.
 - generally involves the use of trial and error.

Exponential Smoothing₁

An exponential smoothing approach that is based on **the previous forecast plus a percentage of the forecast error**

- **Equation:**

$$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$$

where

F_t = Forecast for period t

F_{t-1} = Forecast for the previous period

A_{t-1} = Actual demand or sales for the previous period

α = Smoothing constant

Exponential Smoothing₂

- **Practice:**

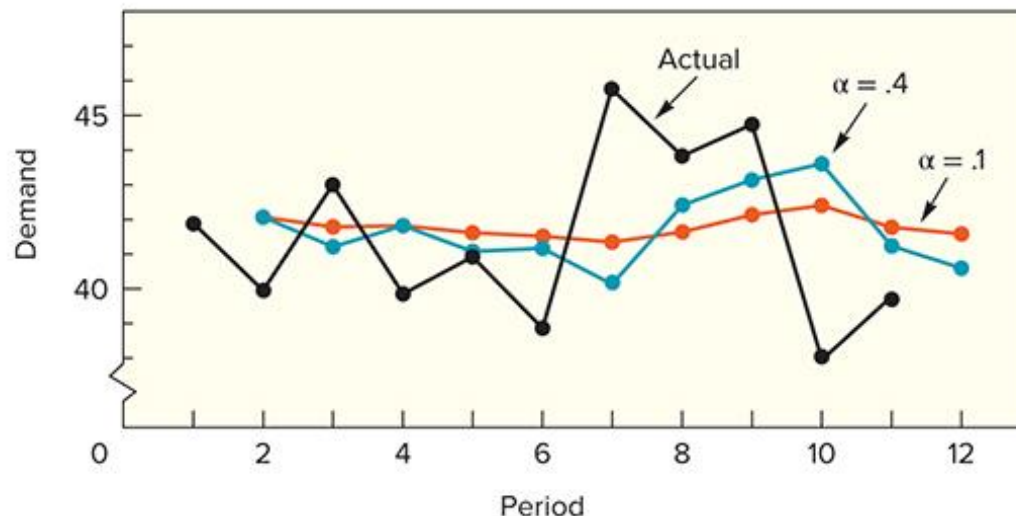
- a) The previous forecast was **42** units, actual demand was **40** units, and $\alpha = 0.10$, compute the new forecast.
- b) According to the new forecast as in part a), if the next new actual demand turns out to be **39**, compute the next forecast.

Period (t)	Actual Demand	$\alpha = .10$ Forecast
1	42	—
2	40	42
3	43	
4	40	
5	41	
6		

starting forecast →

Exponential Smoothing₃

- **Smoothing constant (α)**
 - **Low α values** are used when the underlying average tends to be stable. **Closer its value is to 0, the slower the forecast will be to adjust** to forecast errors.
 - Conversely, **Higher α values** are used when the underlying average change frequently. **Closer the value of is to 1, the greater the responsiveness** and the less the smoothing.

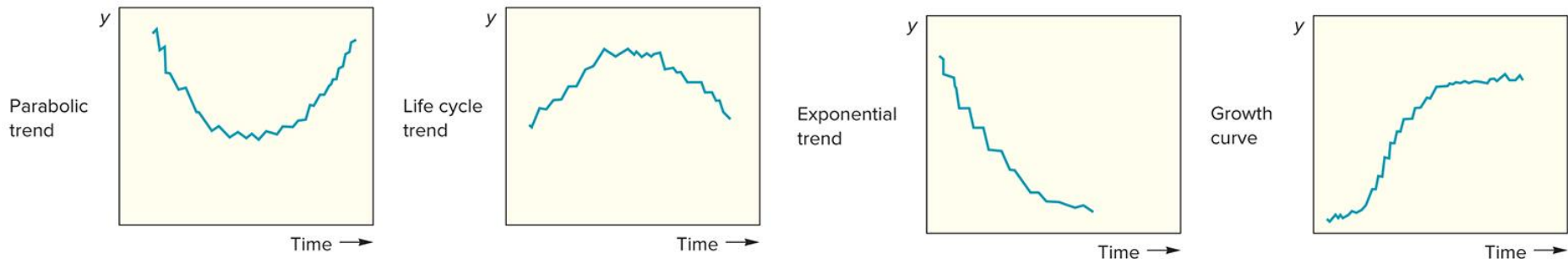


Exponential Smoothing₄

- Exponential smoothing is one of the most widely used techniques in forecasting, because:
 - ease of calculation
 - the weighting scheme can be easily changed

Techniques for Trend

Analysis of trend involves developing an equation that suitably describes trend of data. The trend may be linear, or it may not. Commonly nonlinear trend types:



Techniques

- **Linear trend equation**

Linear Trend Equation₁

Linear Trend Equation can be used to develop forecasts when trend is present.

- ***Equation:***

$$F_t = a + bt$$

where

F_t = Forecast for period t

a = Value of F_t at $t = 0$

b = Slope of the line

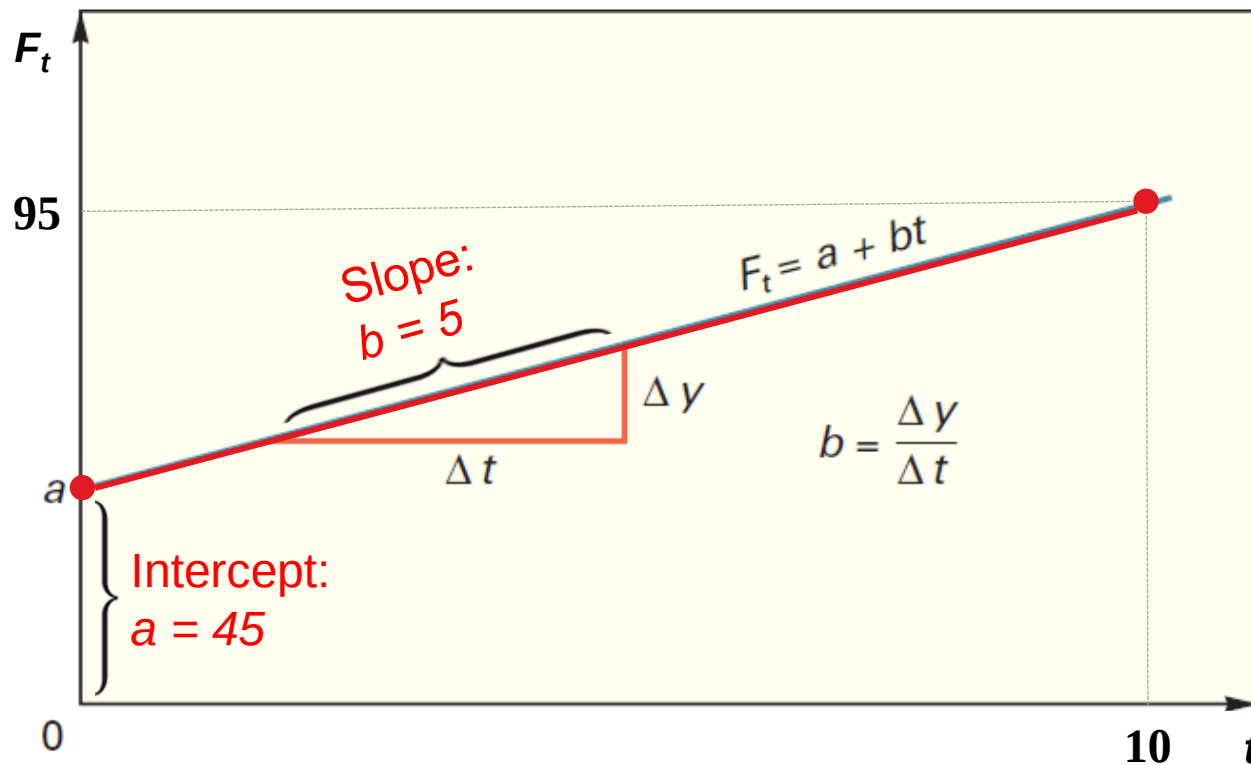
t = Specified number of time periods from $t = 0$

Linear Trend Equation₂

The equation can be plotted by finding two points on the line.

Example:

- Trend equation $F_t = 45 + 5t$; If $t = 10$, the forecast, F_t were be?



Linear Trend Equation₃

- **Practice 1:**

Following table is a cellphone sales for a California-based firm over the last **10** weeks. According to the calculation, ***slope = 7.51*** and ***intercept = 699.4***.

- a) Determine the equation of the trend line.
- b) Predict sales for weeks 11 and 12.

Week	Unit Sales
1	700
2	724
3	720
4	728
5	740
6	742
7	758
8	750
9	770
10	775

Linear Trend Equation₅

The coefficients of the line, a and b , are based on the following two equations.

- ***Equation:***

$$b = \frac{n \sum tY - \sum t \sum Y}{n \sum t^2 - (\sum t)^2}$$

$$a = \frac{\sum Y - b \sum t}{n}$$

where

Y = Value of the time series

n = Number of periods

Linear Trend Equation₆

- **Practice 2:**

- a) Develop a linear trend equation for the following data.
- b) Use the equation to predict the next two values of the series.

Period (<i>t</i>)	Demand (<i>Y</i>)
1	19
2	18
3	15
4	20
5	18
6	22
7	20

Linear Trend Equation₇

- **Practice 2:**

- a) Develop a linear trend equation for the following data.
- b) Use the equation to predict the next two values of the series.

Period (<i>t</i>)	<i>t</i> ²	Demand (<i>Y</i>)	<i>tY</i>
1		19	
2		18	
3		15	
4		20	
5		18	
6		22	
7		20	

Techniques for Seasonality₁

Seasonal variations in time-series data are regularly repeating upward or downward movements in series values that can be tied to recurring events.

Seasonality may refer to regular annual variations.

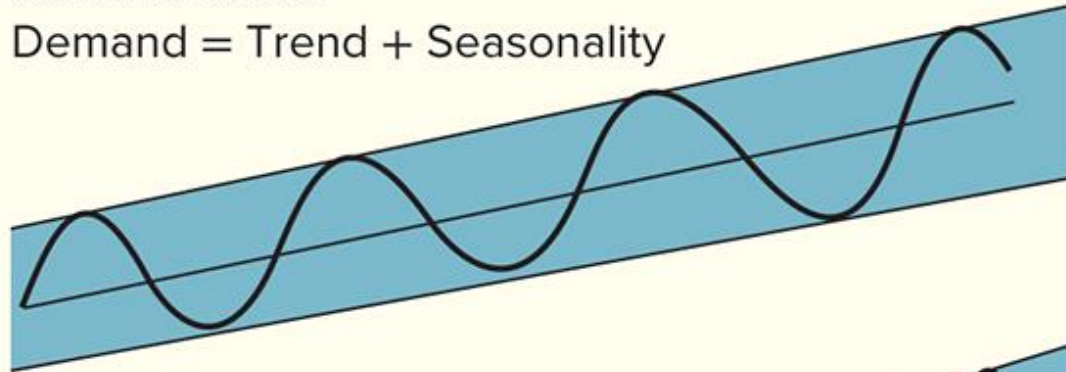
- Models of seasonality
 - **Additive**
 - Seasonality is expressed as a **quantity (e.g., 20 units)** that gets added to or subtracted from the time-series average in order to incorporate seasonality.
 - **Multiplicative**
 - Seasonality is expressed as a **percentage of the average (or trend) amount (e.g., 1.10)** which is then used to multiply the value of a series in order to incorporate seasonality.

Techniques for Seasonality₂

Demand

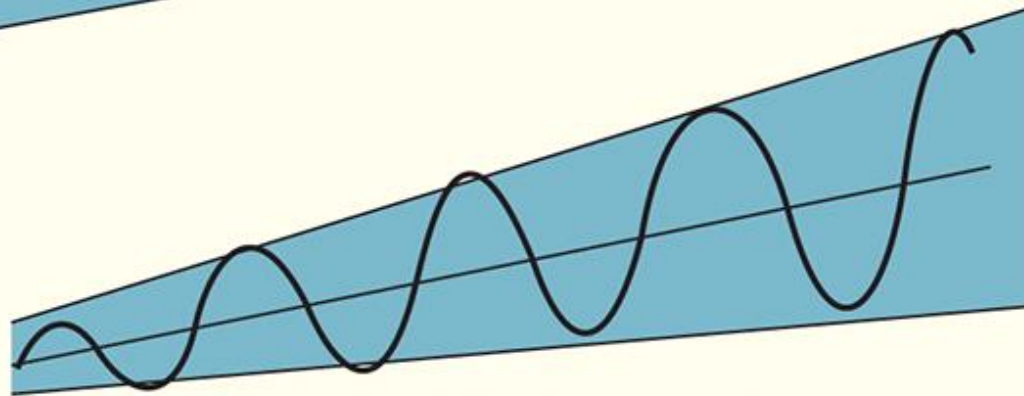
Additive model

$$\text{Demand} = \text{Trend} + \text{Seasonality}$$



Multiplicative model

$$\text{Demand} = \text{Trend} \times \text{Seasonality}$$



Time

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Techniques for Seasonality₃

Seasonal Relatives

- The seasonal percentage used in the multiplicative seasonally forecasting model.

Seasonal relatives are used in one of two ways:

- *To deseasonalize data*
 - *to remove the seasonal component* from the data in order to get a clearer picture of the non-seasonal components of the data series.
- *To incorporate seasonality in a forecast*
 - *Incorporating seasonality* in a forecast is useful when demand has both trend (or average) and seasonal components.

Techniques for Seasonality₄

- **Practice 1:**

A coffee shop wants to estimate demand for the next quarters for hot chocolate. Sales data consist of trend and seasonality.

- a) Seasonal relatives are **$Q1 = 1.20$** , **$Q2 = 1.10$** , **$Q3 = 0.75$** , and **$Q4 = 0.95$** . Use this information to deseasonalize sales for period 1 through 4.

Period	Quarter	Seasonal relatives	Sales (gal.)	Deseasonalized sales (gal.)
1	Q1	1.20	158.4	
2	Q2	1.10	153.0	
3	Q3	0.75	110.0	
4	Q4	0.95	146.3	

Techniques for Seasonality₄

- **Practice 1:**

A coffee shop wants to estimate demand for the next quarters for hot chocolate. Sales data consist of trend and seasonality.

- b) Use appropriate values of seasonal relatives and the trend equation $F_t = 124 + 7.5t$ to estimate demand for periods 5 and 6.

Period	Quarter	Seasonal relatives	Trend sales (gal.)	Forecast sales with seasonality (gal.)
5	Q1			
6	Q2			

Computing Seasonal Relatives

- **Simple Average (SA) Method** can simple to compute the seasonal relatives.
- Seasonal relative of each quarter is the average for that season divided by the average for all seasons.
- ***Equation:***

$$\text{Seasonal relative} = \frac{\text{Particular Season Average}}{\text{Average for All Seasons}}$$

Computing Seasonal Relatives

- ***Practice 2:***

Quarter	Year 1	Year 2	Year 3	Quarter Total	Quarter Average	Quarter Relative
1	20	23	17			
2	10	12	8			
3	25	17	22			
4	28	26	30			

Forecast Accuracy and Control ₁

- ***Accuracy*** and ***control*** of forecasts is a vital aspect of forecasting, so forecasters want to *minimize forecast errors*.
- If errors fall beyond acceptable bounds, *corrective action* may be necessary.
- It is important to determine whether forecasts are performing to an *acceptable level*.

Forecast Accuracy and Control ₂

- Sources of forecast errors including:
 - Omission of an important variable
 - A change or shift in variable the model cannot handle
 - The appearance of a new variable
 - Irregular variations may have occurred
 - Random variation

Forecast Accuracy and Control ₃

Forecast error is the difference between *the value that actually occurred* and *the value that was predicted* for a specific time period.

- **Equation:**

$$\text{Error} = \text{Actual} - \text{Forecast} \quad (e_t = A_t - F_t);$$

where

t = Any given time period

- **Positive errors** result when the forecast is too low
- **Negative errors** result when the forecast is too high.

Forecast Accuracy and Control ⁴

Forecast accuracy is a significant factor when deciding among forecasting alternatives.

Three commonly used measures for summarizing *historical errors* are:

- **Mean absolute deviation (MAD)**
- **Mean squared error (MSE)**
- **Mean absolute percent error (MAPE)**

Forecast Accuracy and Control ₅

$$\text{MAD} = \frac{\sum |\text{Actual}_t - \text{Forecast}_t|}{n}$$

MAD weights errors according to their **absolute value**

$$\text{MSE} = \frac{\sum (\text{Actual}_t - \text{Forecast}_t)^2}{n - 1}$$

MSE weights errors according to their **squared values**

$$\text{MAPE} = \frac{\sum \frac{|\text{Actual}_t - \text{Forecast}_t|}{\text{Actual}_t}}{n} \times 100$$

MAPE weights errors according to **relative error**

Forecast Accuracy and Control ₆

- Practice:**

Compute MAD, MSE, and MAPE for the following data, showing actual and forecasted numbers of accounts serviced.

Period	Actual	Forecast	(A-F) Error	Error	Error ²	(Error /Actual) *100%
1	217	215				
2	213	216				
3	216	215				
4	210	214				
5	213	211				
6	219	214				
7	216	217				
8	212	216				

Forecast Accuracy and Control ₆

- Solution:***

Compute MAD, MSE, and MAPE for the following data, showing actual and forecasted numbers of accounts serviced.

Period	Actual	Forecast	(A-F) Error	Error	Error ²	(Error /Actual) *100%
1	217	215	2	2	4	0.92%
2	213	216	-3	3	9	1.41%
3	216	215	1	1	1	0.46%
4	210	214	-4	4	16	1.90%
5	213	211	2	2	4	0.94%
6	219	214	5	5	25	2.28%
7	216	217	-1	1	1	0.46%
8	212	216	-4	4	16	1.89%
				22	76	10.26%

Forecast Accuracy and Control ₇

- Solution:***

Number of periods	Error	Error ²	(Error /Actual) * 100%
8	22	76	10.26%

$$\text{MAD} = \frac{\sum |e|}{n} = \frac{22}{8} = 2.75$$

$$\text{MSE} = \frac{\sum e^2}{n - 1} = \frac{76}{8 - 1} = 10.86$$

$$\text{MAPE} = \frac{\sum \left[\frac{|e|}{\text{Actual}} \times 100 \right]}{n} = \frac{10.26\%}{8} = 1.28\%$$

Forecast Accuracy and Control ₈

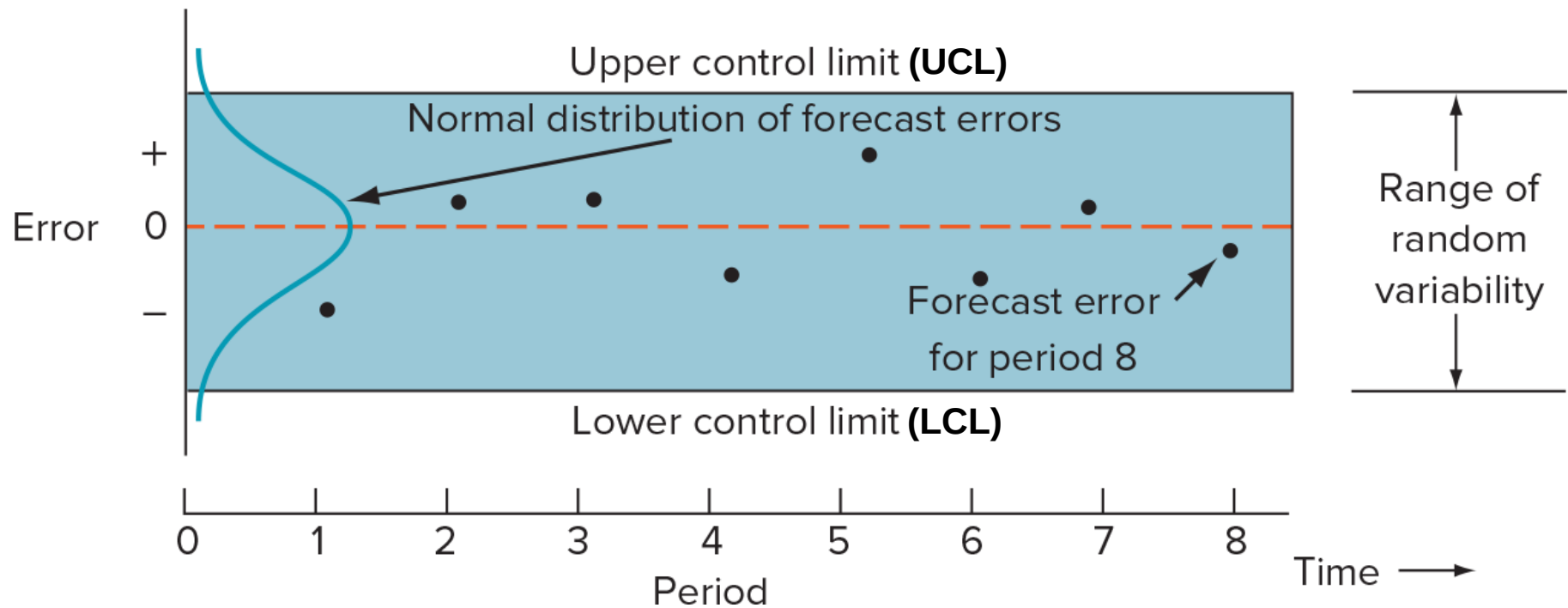
- **Mean absolute deviation (MAD)**
 - The easiest to compute, but weights errors linearly.
- **Mean squared error (MSE)**
 - Through squares errors, thereby giving more weight to larger errors, but typically cause more problems.
- **Mean absolute percent error (MAPE)**
 - Should be used when errors need to be measured in relative views.

Monitoring the Forecast

- Tracking forecast errors and analyzing them can provide useful insight into whether forecasts are performing satisfactorily.
- **Control charts** are useful for identifying the presence of *non-random error* in forecasts.

Control Chart

- In control chart, the point of forecast error that fall within the two control limits, represent an acceptable ***margin of error***.



Control Chart Construction

Step 1. Compute the ***error metrics*** (i.e. MAD, MSE or MAPE).

Step 2. Estimate ***standard deviation*** of the distribution of errors:

$$\sigma = \sqrt{\text{error metrics}}$$

Step 3. Obtain the UCL and the LCL:

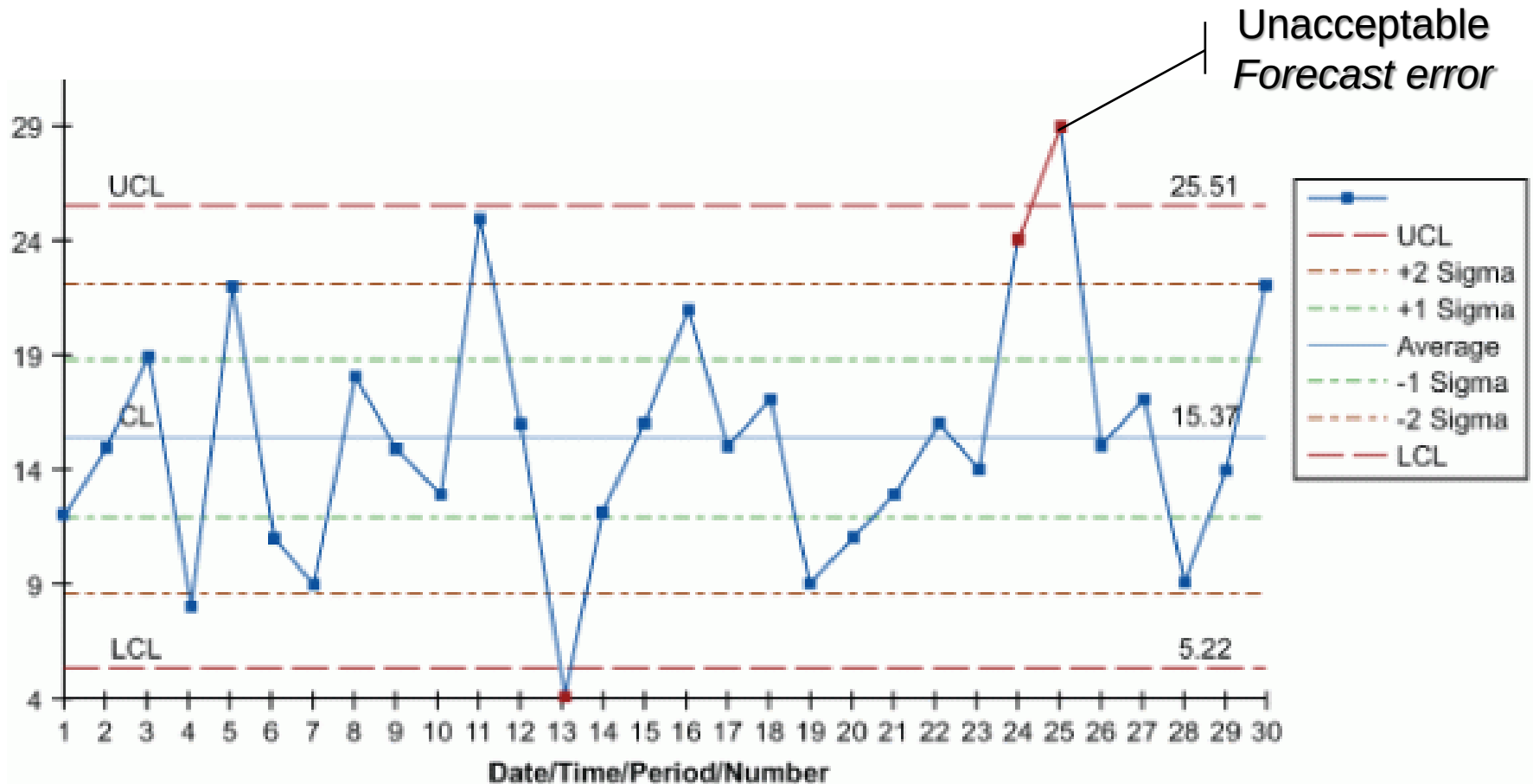
$$UCL = 0 + z\sigma$$

$$LCL = 0 - z\sigma$$

where

z = Number of standard deviations

Standard Deviation



Control Chart Construction

- **Practice:**

According to following data and each *error metric* (i.e. MAD, MSE, MAPE), compute **2σ** control limits for forecast errors.

Period	Actual	Forecast	Error metric
1	217	215	• <i>MAD</i> = 2.75 • <i>MSE</i> = 10.86 • <i>MAPE</i> = 1.28%
2	213	216	
3	216	215	
4	210	214	
5	213	211	
6	219	214	
7	216	217	
8	212	216	

Control Chart Construction

- Solution:**

$$\sigma = \sqrt{MAD} = 1.66$$

$$UCL = 0 + 2 * 1.66 = 3.32$$

$$LCL = 0 - 2 * 1.66 = -3.32$$

$$\sigma = \sqrt{MSE} = 3.30$$

$$UCL = 0 + 2 * 3.30 = 6.59$$

$$LCL = 0 - 2 * 3.30 = -6.59$$

$$\sigma = \sqrt{MAPE} = 1.13\%$$

$$UCL = 0 + 2 * 1.13\% = 2.26\%$$

$$LCL = 0 - 2 * 1.13\% = -2.26\%$$

Period	Actual	Forecast	(A-F) Error
1	217	215	2
2	213	216	-3
3	216	215	1
4	210	214	-4
5	213	211	2
6	219	214	5
7	216	217	-1
8	212	216	-4
AVE.	214.5	214.75	-2

Choosing a Forecasting Technique

When selecting a forecasting technique, decision maker must take a number of factors into consideration:

- Cost
- Accuracy
- Availability of historical data
- Availability of forecasting software
- Time needed to gather data and to prepare a forecast
- Forecast horizon

Using Forecast Information

- Manager can take a reactive or proactive approach to a forecast.
- **Reactive approach**
 - Views forecast as probably future demand, and reacts to that demand
- **Proactive approach**
 - Seeks to positively influence demand, and by means of many strategies

Operations Strategy

The better forecasts are, the more able organizations will be to take advantage of future *opportunities* and reduce *potential risks*.

- A worthwhile strategy is to work to improve *short-term* forecasts.
- Accurate *up-to-date information* can have a significant effect on forecast accuracy, include:
 - Prices
 - Demand
 - Production rates
 - Reduce the time horizon forecasts have to cover
- Sharing forecasts or demand data through the *supply chain* can improve forecast quality and increase operations performance.

Summary

- Forecasts are vital inputs for the design and the operation of the productive systems because they help managers to anticipate the future.
- Forecasting techniques can be classified as qualitative or quantitative.
- Qualitative techniques rely on judgment, experience, and expertise to formulate forecasts.
- Quantitative techniques rely on the use of historical data or associations among variables to develop forecasts.

Summary

- Some forecasting techniques better than others, but no technique works all the time.
- Moreover, all forecasts include a certain degree of inaccuracy, therefore, it is important to provide a measure of accuracy.
- When selecting a forecasting technique, a manager must choose a technique that will serve the intended purpose at an acceptable level of cost and accuracy.



The End